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HANDMADE MANDOLIN, PART 1: RED-SPRUCE TOP

From "Handmade Music"
episode DHMM-105 -- More Projects »

The quest to build the perfect mandolin mandolin leads DIY's *Handmade Music* to the Ryman Auditorium in Nashville, Tennessee. It was from the Ryman's stage that a man named Bill Monroe introduced the world to bluegrass music. That was 1945 and it was an F5 style mandolin that delivered the unique sound that has become so familiar to fans of bluegrass and traditional American music. Since that first introduction, the world hasn't stopped tapping its toe to that high lonesome sound, and musicians haven't stopped trying to replicate Mr. Monroe's style. But 1920s-era Gibson F5s are hard to come by, and that's where Lynn Dudenbostel comes in. Lynn is a craftsman who replicates the look and sound of vintage mandolins like those built by the masters during the golden era of guitars and mandolins. In the first of three episodes devoted to the mandolin, we learn why the instrument is special and witness the first steps in the process of building one from scratch.

Like the acoustic guitars made before World War II, classic mandolins have tops made from prized red-spruce wood.

In this first segment, we begin with the construction of a red-spruce top. The top features a set of tone bars on the inside of the mandolin and Lynn shows why this step is referred to as "giving the instrument voice". Old-school gives way to high-tech when Lynn introduces the use of a computerized router to carve the top and the curly maple back.

In later segments, Lynn prepares the back and sides of the instrument from curly maple, and viewers get a taste of how Dudenbostel mandolins sound when pushed with a virtuoso performance. Andy Leftwich and his fellow Kentucky Thunder band-mates put a Dudenbostel mandolin through its paces in a bluegrass jam.

The "Holy Grail" of mandolins -- an original Gibson F5.

Materials:

Red-spruce stock
CNC (computer-driven) router
Jointer
Drum sander (thickness sander)
Hot hide glue
Gluing fixture
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Red-Spruce Mandolin Top

Master luthier Lynn Dudenbostel builds stringed instruments by hand, in the style of masters who made the great instruments in earlier decades. He built his reputation predominantly by replicating a specific classic instrument -- the Gibson F5 mandolin. "1922 through 1924 was considered the golden era for Gibson mandolins," says Dudenbostel, "and Lloyd Loar was the acoustic engineer at Gibson during that time, and those are the most sought-after mandolins." In fact, some of Loar's original hand-written documentation has been located which specifies the Virginia Adirondack spruce (a.k.a., the red spruce) as the wood of choice for the instrument's top. So the objective for this *Handmade Music* series is to build a classic F-5 style mandolin using a top made from red spruce just as acoustic engineer Lloyd Loar would have wanted.

Figure A

- In selecting red-spruce stock for building a mandolin, Dudenbostel looks for evenness of grain, good definition between the winter and summer growth in the wood (seen in cross-section, like the familiar rings of a cut tree) and a ringing **tap tone** (as opposed to a dull "thud") when the piece of wood is lightly struck (**figure A**).

- Once the wood has been selected and cut to size, two pieces jointed for creation of the top. Typically, two book-matched pieces of spruce (**figure B**) are joined together by their jointed faces to create the single top-piece.

Once the joining surfaces have been jointed smooth, they are glued together using **hot hide glue** and a gluing fixture. Hide glue is made from ground animal hides and is ideally suited for soft-wood applications.

To process the wood to make it ready for carving, one side -- which will eventually be the bottom surface of the top piece -- must be made perfectly flat. The face that will eventually be the top begins as a wedge shape. To flatten the bottom surface, the glued pieces are passed through the thickness sander. Two temporary rails balance the wedge shape as it passes through the sander (**figure C**).

Figure B

Figure C

The planed piece is left fairly thick as the extra thickness is needed for the arch-top style of the Gibson F5. Rather than bending thin stock to shape, Lynn creates the arch by carving the shape from thicker stock. A template is used to mark the outline of the top onto the stock (**figure D**).

Carving the thick top was once a process that had to be done by hand, but Lynn makes use of some modern technology for this phase of the mandolin's construction. After the book-matched pair has been glued and sanded flat on one surface, the stock is then placed on a CNC machine to do the rough carving (**figure E**).

Figure D

Figure E

CNC stands for **computer numerical control**, and the CNC machine is actually a router that is driven by a computer. In Lynn's case, the mandolin's top will be finished by hand, but he does use the CNC router to rough-carve the basic shape. Later, he Lynn will do the final carving, graduation and "voicing" of the instrument by hand. Rough-carving begins with the top's inner surface as surplus wood is dished out (**figure F**). The top is then flipped over to expose the outer surface. Once the computer has completed its program, the basic shape of the mandolin top is visible. This CNC process takes slightly less than 2 hours.

Figure F

Lynn points out that using the CNC for rough carving isn't straying too far from the spirit of the classic instrument-making since the highly desirable instruments made in the early 1920s were made using a mechanical duplicator -- the latest technology available at that time. "I take a pattern that I've carved by hand," says Dudenbostel, "and I have a digitizing program that collects x, y and z data-points from that pattern, and converts them into a program to rough-cut the top and back." The final and detailed carving is still done by hand.

In the segment that follows, Lynn does just that. He hand-shapes the top and back.

RESOURCES:

Stewart MacDonald's Luthier Supply Shop
Website: www.stewmac.com

Luthier's Mercantile International, Inc.
Website: www.lmii.com

Pioneer Valley Luthier Supply Company
Website: www.pioneervalleyluthier.com

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

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HANDMADE MANDOLIN, PART 2: HAND-SHAPING THE TOP AND BACK

From "Handmade Music"
episode DHMM-105 -- More Projects »

In this segment of DIY's *Handmade Music*, luthier Lynn Dudenbostel hand-shapes the mandolin top and back after they have been rough-cut using a computerized router known as a CNC machine.

From this stage at Nashville's Ryman Auditorium, Bill Monroe introduced the world to modern bluegrass. He played the fabled Gibson F5 mandolin, the model on which this modern replica was based.

Materials:

Red-spruce stock
CNC (computer-driven) router
Band saw
Finger planes
Hand scrapers
Sandpaper
Violin caliper
Hot-hide glue
Gauze reinforcement
Gluing fixture
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Hand-Shaping the Top

After some additional trimming with a band saw, Lynn reverts to the old-school techniques using finger planes (**figures A and B**) and other specialty tools. Finger-planes are tiny, specialized planes that are also used by violin-makers.

Figure A

Figure B

As he hand-shapes the top using the finger planes, he checks his progress periodically with a violin caliper (**figure C**) which measures the precise thickness of the plate.

As he carves the plate to specific dimensions, he also tap-tests the wood occasionally to check its sound. "There will be variation [in the sound]," says Dudenbostel, "depending on the hardness of the wood, the stiffness. Each wood calls for individual characteristics in carving. That's why you can't just machine each one on a CNC to a given dimension. There still has to be that human input in carving each one.

Figure C

Once the hand-carving of the top reaches a certain point, it's taken back to the CNC router (**figure D**) to create the **F-holes**. These ornately shaped openings if the mandolin's top (**figure E**) are its sound holes.

Figure D

Figure E

Lynn describes the essential function of the sound holes in the mandolin. "Without a hole in the top of the instrument," says Dudenbostel, "there's really no way for that body to project sound. You can think of the body of a mandolin or a guitar as an air-pump -- because any time you strum the strings the top vibrates, and there's air moving in and out of that sound hole. Without that, you'd have a very quiet instrument." For this process, computer precision of the CNC is a great asset because, while F-Holes enhance the look of the instrument (**figures F and G**), they are critical to the sound. There is little room for error.

Figure F

Figure G

At this point, the top may look complete but it's not. The tone bars on the inside of the mandolin are created and installed next (**figures H and I**). Though they may look like braces, the mandolin doesn't require additional internal support; just acoustic enhancement. "In an arch-top style mandolin, the bridge can support the weight of the strings and the tension," says Dudenbostel, "without any additional support. The tone bars inside the instrument are just for transmitting the vibrations throughout the top."

Figure H

Figure I

"Fitting the tone bars," he continues, "can be a fairly lengthy process because you have a very complex curve on the inside of the instrument. That has to be fitted by hand. I'll use finger planes, scrapers, sandpaper -- whatever's necessary. And it has to be an absolutely perfect fit." Once the tone bars have been trimmed to their final form and fitted, they are carefully glued in place using hot hide-glue (**figures J and K**).

Figure J

Figure K

While the tone bars provide no support, the gauze strips over the F-holes (**figure L**) do provide extra strength. Lynn will cut away the gauze over the holes but the edges are now reinforced and that should prevent any splitting through the years.

Figure L

It's at this point that Lynn truly creates the sound of this mandolin. As he says, he is "giving the instrument its voice" as he carves the tone bars to

specific dimensions (**figures M and N**). The dimensions and the carving techniques are gleaned from years of research. Lynn listens to the mandolins of yesteryear and, from their voices, he learns styles and techniques worthy of emulation. Every shaving removed from a tone bar affects the acoustics. This is deliberate and thoughtful work, and is the final step in preparation of the top.

Figure M

Figure N

- With the creation of the top complete, focus shifts to creating a back and later the sides for the mandolin. First, a single solid block of curly maple is used for the instrument's back (**figure O**). The stock must be thick enough to allow the deep carving required to create the curves of an F-5.
- After leveling the piece with a thickness sander, Lynn returns to the CNC machine and carves a rough approximation of the mandolin's back. Again, the machine is simply duplicating a shape chosen and programmed by Lynn. When the inner and outer profiles are carved, the shape is trimmed at the band saw (**figure P**).
- Once trimmed to shape, Lynn sands the outer portion of the back. The inner carving of the back must be done by hand and, like the carving of the top, the work is done with the finger planes normally used by violinmakers (**figure Q**).
- Lynn scrapes away the maple little by little until he hears what he wants to hear when he conducts his tap-test. Exactly what he's listening for is of those unwritten, elusive things that only a luthier with years of experience can know. As Lynn says, "When it's right, it's right."

Figure O

Figure P

Figure Q

In the segment that follows, Lynn creates the mandolin's curved sides.

RESOURCES:

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Website: www.pioneervalleyluthier.com

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

- **ALSO IN THIS EPISODE:**
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- Handmade Mandolin, Part 2: Hand-Shaping the Top and Back
- Handmade Mandolin, Part 3: Creating the Curved Sides
- Handmade Mandolin, Part 4: Riser Block
- WEB EXTRA: *Handmade Music* Song Downloads

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HANDMADE MANDOLIN, PART 3: CREATING THE CURVED SIDES

From "Handmade Music"
episode DHMM-105 -- More Projects »

In this episode of DIY's *Handmade Music*, Lynn Dudenbostel continues the process of creating an instrument that looks and sounds like a vintage F-5 style mandolin. In this segment he focuses on the body of the instrument. Earlier Lynn crafted a top from two joined pieces of red spruce and then carved a back from one solid piece of curly maple. Now his focus shifts to the sides that are formed by bending curly-maple strips.

Lynn Dudenbostel creates the curved sides for the mandolin using a custom-made bending form.

Andy Leftwich of the band Kentucky Thunder provides a worthy workout for a newly minted Dudenbostel mandolin.

Materials:

Curly-maple stock
Mahogany stock
Band saw
Drum sander (or thickness sander)
Bending device (custom-made using metal pipe and propane torch)
Bending form
Instrument-maker's glue
Straight-edge
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Creating the Curved Sides

- Strips of curly maple are cut to size on a band saw, then brought to the proper thickness using a thickness sander.
- Some curves of the mandolins sides are extreme and some are subtle. The bending is aided by a device that uses a propane torch to heat a metal pipe. The wood strips are moistened, then gradually bent over the hot pipe. The hot metal surface turns the water in the wood to steam, allowing the wood to bend more easily (**figures A and B**). Lynn designed this device in a way that he can bend the wood on the hot metal while keeping his hands a safe distance from the heat.

Figure A

Figure B

- The goal in this phase of the construction is to bend three strips to form the perimeter of the mandolin. It's a slow, gradual process, and a template is used as a guide (**figure C**).
- The extreme curves of a mandolin make this a tricky process. Ultimately the strips will fit inside a form that will hold the sides together while they cool. As he works bending the strips of wood, Lynn references the form as a guide as well (**figure D**). Forms like the one Lynn uses can be purchased from luthier-supply shops, but Lynn built his forms by hand.

Figure C

Figure D

Each side piece receives equal attention in the bending process (**figure E**).

After a few hours of work Lynn has formed all the needed side elements (**figure F**). "You put the pieces in the form," says Lynn "and hold them in the form until they dry and have cooled off -- which is normally just a few hours. Then they normally take a set to that shape."

Figure E

Figure F

With the side elements bent to their proper shapes, Lynn begins work creating and gluing in the **head-block, tail-block** and **point blocks**. These interior parts are made from solid mahogany and cut for precise fit using the band saw. The use of mahogany for these parts is based on the traditions of the original instrument makers of the 1920s. Templates culled from vintage instruments are used to create these precisely sculpted pieces (**figure G**).

Figure G

Once the pieces are all cut to the proper size and shape, Lynn begins gluing them in place around the rim he's created using instrument-maker's glue. The head block goes on the inside of the rim on the upper side where the neck will connect later (**figure H**).

The tail block is glued to the lower part of the instrument, and the point blocks fit inside the triangular shapes created by the joined strips of maple (**figure I**).

Figure H

Figure I

After the glue is dry Lynn can remove the rim for the first time and see the shape of his newest creation (**figure J**) .

There's some sanding to be done (**figure K**)and a little more essential crafting.

Figure J

Figure K

- Once the sanding is complete, Lynn installs **kerfed lining** around the rim (**figure L**). "Kerfed lining is a small, flexible strip [of wood] that has saw-kerfs cut in it," says Dudenbostel, "so that it can make those real tight bends on the inside of the instrument. It's glued on the rim where the top and back will join together to give you more glue surface-area. When you put the binding on an instrument, you actually rout away most of the glue joint between the sides and top or back. You need something else there to give you more glue surface. That's where the kerfed lining comes in."
- After removing the pins that serve as clamps for the delicate kerfed lining (**figure M**), Lynn checks and removes any excess glue.

Figure L

Figure M

- Finally, once the pins have been removed, Lynn sands the edges even with the sides (**figure N**).

Figure N

To this point, Lynn Dudenbostel has put in nearly two weeks work just to complete the top, back and sides of the mandolin. In the segment that follows, Lynn completes this phase of the mandolin construction by installing the riser block. In addition, bluegrass band Kentucky Thunder demonstrate exactly what a Dudenbostel mandolin can sound like at a full gallop.

On the stage at the world-famous Ryman Auditorium in Nashville, host Jeff Wilson holds the instrument that introduced the world to the unmistakable bluegrass sound -- the Gibson F5 style mandolin.

RESOURCES:

Stewart MacDonald's Luthier Supply Shop
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Luthier's Mercantile International, Inc.
Website: www.lmii.com

Pioneer Valley Luthier Supply Company
Website: www.pioneervalleyluthier.com

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

ALSO IN THIS EPISODE:

- Handmade Mandolin, Part 1: Red-Spruce Top
- Handmade Mandolin, Part 2: Hand-Shaping the Top and Back
- Handmade Mandolin, Part 3: Creating the Curved Sides
- Handmade Mandolin, Part 4: Riser Block
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HANDMADE MANDOLIN, PART 4: RISER BLOCK

From "Handmade Music"
episode DHMM-105 -- More Projects »

In this segment, Lynn comes one step closer to completion of the mandolin's body by creating and installing the riser block. Later in the segment, bluegrass band Kentucky Thunder demonstrates exactly what a Dudenbostel mandolin can sound like at a full gallop.

Lynn Dudenbostel dry-fits the mandolin's body as he adds creates riser block.

Materials:

Maple stock
Tortoise-colored plastic (available from luthier supply-shops)
Drum sander
Band saw
Instrument maker's glue
Straight-edge
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Riser Block

Thus far in the process, Lynn Dudenbostel has built the mandolin's body. The body's major components are the top (complete with tone bars), the maple back, and rim assembly -- which consists of a maple rim, head-block, tail-block and point-blocks. The internal blocks help the instrument hold its shape and are made of mahogany. With just a few more enhancements, he will be finished with this first major step of construction.

Lynn adds a riser block to the top of the rim (**figures A and B**). Eventually the riser will hold the mandolin's neck. Because the neck will slope back, away from the top, the riser block has to slope down at the same angle. In this case the angle is about five and a half degrees. The riser block will also eventually have a dovetail cut into it, so Lynn uses maple for added strength.

Figure A

Figure B

Also, for decorative purposes Lynn trims the tips of the point blocks. He replaces the wood with tortoise-colored celluloid plastic and shapes it using a drum sander (**figure C**). This is strictly a decorative touch and it's also in keeping with the tradition of F-5 mandolins (**figure D**).

Figure C

Figure D

- With the rim essentially complete, Lynn positions the top temporarily in place. Using the top as a guide, he glues the riser block to the mahogany head block (**figure E**).
- After the hide-glue dries, Lynn uses a template to mark the dovetail into the riser block. Later he can cut the dovetail with a band saw and create the joint that will eventually accept the mandolin's neck.

Figure E

Kentucky Thunder in a full-tilt bluegrass workout.

As he nears completion of the mandolin's body, Lynn Dudenbostel comments on the feeling that can come from hand-crafting a musical instrument. "To hear somebody performing with your instrument is a real charge," says Dudenbostel. "I remember the first time I ever heard one of my instruments on the radio, I almost ran my car off the road. It's tremendously exciting, and it doesn't matter if it's somebody playing professionally. A good portion of my customers are just individuals who play for the pure enjoyment of it -- or, like me, for the annoyance of their family!"

Despite what his family may think of the sound of Lynn's instruments, the members of Kentucky Thunder have no problem at all with it. Andy Leftwich and his band-mates treated viewers to a sample of what a Dudenbostel mandolin can sound like in as part of a first-rate bluegrass ensemble.

In the episode that follows, the mandolin's back and the sides are ready to be joined to form the finished body.

RESOURCES:

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Website: www.lmii.com

Pioneer Valley Luthier Supply Company
Website: www.pioneervalleyluthier.com

Kentucky Thunder
For more information on the band that appeared in DIY's *Handmade Music*, Kentucky Thunder -- and to hear streaming audio and download samples of their music -- visit the website for Skaggs Family Records, www.skaggsfamilyrecords.com.

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

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HANDMADE MANDOLIN, PART 5: ASSEMBLING THE BODY

From "Handmade Music"
episode DHMM-106 -- More Projects »

This episode of DIY's *Handmade Music* continues our three-part series on the building of a great custom mandolin. Luthier Lynn Dudenbostel glues the top and the back to the sides and then adds binding to complete the body. Finally, veiwers learn the first steps of making a neck.

The Gibson F5 mandolin, on which this handmade replica is based, featured a top made of red-spruce. Modern luthier's like Lynn Dudenbostel strive to replicate the look and, more importantly, the sound of the great mandolins from the past.

In this segment, the mandolin's body gets assembled.

Materials:

Instrument maker's glue
Clamps
Table router
Sandpaper
Wooden dowel
Finger planes
Gouges
Sharp chisel
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Gluing the Top and Back to the Sides

In the previous episode, Lynn Dudenbostel completed the construction of the red spruce top, crafting it with a combination of old and new technology. A computer-aided router cut the shape close to vintage specs, and Lynn, like luthiers of old, carved the final graduations using finger planes. The back was carved the same way, but Lynn used a block of curly maple. Finally he employed more "old world technology" to bend strips of maple around a heated pipe. These strips (or sides) formed the shape of the mandolin's perimeter. Now, to finish the body, Lynn will glue the top and the back to the rim.

- The kerfed lining Lynn installed earlier provides plenty of surface area to spread a generous amount of instrument-maker's glue (**figure A**)
- Once the glue is applied, Lynn places the curly maple back in place (**figure B**).

Figure A

Figure B

- Plenty of clamps are used to secure the back to the rim as the glue dries(**figure C**).
- After drying overnight, Lynn autographs his work. It's his last opportunity to work inside the instrument, because after the next step, attaching the top, the interior of the instrument will no longer be accessible.
- Lynn attaches the top just as he did the back the back (**figure D**). He uses the same glue and spreads it along the kerfed lining.

Figure C

Figure D

After at least four hours of drying, the body is removed from its form (**figure E**) and the mandolin has begun to take shape (**figure F**), but there's still plenty of work left to be done. The overlap from the top and back needs to be cleaned up. Also, decorative binding must be installed before this body can be declared complete.

Figure E

Figure F

A quarter-inch flush-trim router bit is used remove most of the overhang from the top and back (**figure G**).

Then 100-grit sandpaper is used smooth out the bumps and clean up the surfaces. Lynn wraps the paper around a wooden dowel to work the bends and curves of the instrument (**figure H**). In time, Lynn finishes with a perfectly unified surface that melds the back, sides and top.

Figure G

Figure H

The biggest challenge of this phase of the mandolin construction is still ahead. The most distinctive characteristic of an F-5 mandolin is the swirling scroll at the top of the body (**figure I**). That feature simply cannot be created using a router bit. Lynn makes use of a variety of hand-ools to form the scroll.

First he carves away excess wood using a razor-edged knife. Then he glues sandpaper to a sliver of plastic and uses that to sand away surplus wood inside the scroll (**figure J**). Finally he uses gouges, chisels and finger-planes to sculpt the look of a classic mandolin scroll. It's this type of work that sets luthiers like Lynn Dudenbostel apart. There are no templates to guide him in this part of the process. His resources are experience and creativity.

Figure I

Figure J

When Lynn is finished with his handcrafting of the scroll, there is a nice separation between the scroll and the rest of the body (**figure K**). In the segment that follows, Lynn adds decorative binding to the outside edge of the instrument.

RESOURCES:

Stewart MacDonald's Luthier Supply Shop
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Pioneer Valley Luthier Supply Company
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Figure K

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

- **ALSO IN THIS EPISODE:**
- Handmade Mandolin, Part 5: Assembling the Body
- Handmade Mandolin, Part 6: Decorative Binding
- Handmade Mandolin, Part 7: Creating the Neck
- Handmade Mandolin, Part 8: Completing the Neck
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HANDMADE MANDOLIN, PART 6: DECORATIVE BINDING

From "Handmade Music"
episode DHMM-106 -- More Projects »

In this segment of DIY's *Handmade Music*, luthier Lynn Dudenbostel adds decorative binding to the outside edge of the instrument. But before he does, he takes the opportunity to talk about an original Gibson F5 that's in his shop for repairs. It's vintage mandolins like the classic Martin F5 one and the luthiers who built them that inspire the design and the techniques used in construction today. Modern makers and repairmen like Lynn Dudenbostel relish every opportunity to study the great ones.

The Gibson F5 is regarded as "the holy grail" of mandolins.

The one shown in the figure at right was built in the 1920s and was originally owned by Earle Taylor who played in a group called The Stony Mountain Boys -- the first bluegrass group to ever play Carnegie Hall. This particular instrument was also reportedly played by the legendary Bill Monroe on several occasions.

The Archetypal Mandolin: The Vintage F5

The original Gibson F5 mandolin featured in this segment was built in the mid-1920s. This particular model is commonly referred to as the "fern" because of the ornate inlay on the head stock. This particular instrument (**figure A**) originally belonged to a famous musician named Earle Taylor who played with a group called The Stony Mountain Boys -- the first bluegrass group to ever play Carnegie Hall. Reportedly, Bill Monroe also played this particular mandolin on several occasions. Despite considerable wear to the body and neck of the instrument (**figure B**), this mandolin maintains a wonderful sound.

Figure A

Figure B

It's mandolins like this that serve as a model and basis for the work that contemporary luthiers like Lynn Dudenbostel do. While this historic instrument is in his shop for repair work, Lynn takes the opportunity to study the various aspects of its construction.

"What's of particular importance to builders," says Dudenbostel "are the top and back thickness measurements." Lynn uses an instrument known as a Hacklinger gauge (**figure C**) to take those measurements. Mapping out these measurements helps provide an understanding of what's necessary to re-create the distinctive sound of these classic instruments. This device has a small magnet that goes inside the instrument.

A scale on the side of the gauge provides readings for the precise thickness measurements, in millimeters, with respect to the instrument body (**figure D**). By taking numerous readings over the top and back of the instrument, luthiers can determine the graduations in thickness that help create the mandolin's sound. Later, they can attempt to recreate those graduations when they build their new instruments.

Figure C

Figure D

The Mandolin's Decorative Binding

Lynn plays and studies an *original* Gibson F5. This particular instrument reportedly was once played by Bill Monroe, the father of modern bluegrass.

Materials:

Table router
Celluloid binding
Instrument-maker's glue
Rotary tool
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Steps:

- Inspired by the classic F5, Lynn can now finish the construction of his mandolin's body by adding celluloid binding to its perimeter. First he must cut a rabbet around the perimeter of the body to accommodate the binding. He measures the binding with a thickness gauge (**figure E**) and uses the binding itself to determine the exact dimension of the rabbet to cut around the perimeter of the body.
- At the router table adjusts the spiral bit and jig to reflect the measurement he comes up with. He then makes the cut in several passes to prevent chipping (**figure F**) The first pass creates a good ridge and after checking the progress he adjusts his fixture slightly and cuts away additional material.

Figure E

Figure F

- When it appears that the channel has been cut to adequate dimensions, Lynn checks it against the binding to ensure a good fit (**figure G**).
- The router can leave little spurs of wood along the edge, so Lynn trims them with a chisel to create a perfectly clean surface along the rabbet (**figure H**).

Figure G

Figure H

- As before, there's one spot a router-bit won't help -- around the scroll. Lynn sketches out a pattern that extends the rabbet around the scroll (**figure I**)
- He then cuts away the excess wood using a rotary tool (**figure J**). A good way to judge the skill of a luthier is check out the scroll. This is where good ones like Lynn earn their money. The precision is achieved freehand, and this technique can only be perfected through skill and experience.

Figure I

Figure J

- After additional cleanup with chisels, Lynn is finally ready to glue binding to the instrument using a two-step process. With super glue, Lynn attaches one layer of binding with a white surface facing out (**figure K**). Typically he starts with the scroll and works out and around the top.
- When finished with the first layer, he adds another layer with a tortoise shell design facing outward (**figure L**).

Figure K

Figure L

- This time he uses a gelatin mixture comprised of acetone and scraps of binding melted together to create a bond. In essence the acetone melts the celluloid material together and permanently bonds the strips. Lynn wraps the binding around the scroll and works down the instrument (**figure M**). He uses a heat gun to soften the binding a make it more pliable.
- Lynn's clamping method for securing the binding is a heavy application of masking tape (**figure N**).

Figure M

Figure N

The tape is removed after at least a day's drying time. Lynn then uses a hand scraper to remove any excess glue and to clean up the edges (**figure O**).

Figure O

RESOURCES:

Stewart MacDonald's Luthier Supply Shop
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Luthier's Mercantile International, Inc.
Website: www.lmii.com

Pioneer Valley Luthier Supply Company
Website: www.pioneervalleyluthier.com

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

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HANDMADE MANDOLIN, PART 7: CREATING THE NECK

From "Handmade Music"
episode DHMM-106 -- More Projects »

In this episode of DIY's *Handmade Music*, Lynn Dudenbostel continues with the creation of a handmade mandolin, built in the tradition of a classic Gibson F5. In this segment, he begins work on creating various elements of the mandolin's neck.

Andy Leftwich of bluegrass band Kentucky Thunder, and mandolin player for Ricky Skaggs, gets to give the Dudenbostel mandolin a trial run. Based on both tone and appearance, Andy considers this "a great mandolin."

Materials:

Cocobolo-wood stock
Tahitian mother-of-pearl
Band saw
Table saw with cross-cut sled
Table router
Template for fret slots
Epoxy glue
Cocobolo sawdust
Brass tubing
Straight-edge
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Creating the Fingerboard

Thus far Lynn has a completed mandolin body, minus the wood finish. It's now time to shift gears and begin creating the mandolin neck. The neck will be constructed from a variety of woods. For this instrument, Lynn opts to veer a bit from the traditional coarse. Where most vintage mandolins have ebony fingerboards, but for unique styling Lynn has chosen **cocobolo** wood. For the sake of matching the look, he sticks with cocobolo for the peg-head veneer.

Though most vintage mandolins have ebony fingerboards, for unique styling Lynn opted for cocobolo wood for the mandolin built in this series.

After roughing-in the fingerboard at the band saw, Lynn tapes a template to the cocobolo stock (**figure A**) and perfects the shape at the router table (**figure B**).

Figure A

Figure B

Then, with a shallow cut at the table saw Lynn uses a template with precut fret placements to add fret slots into the fingerboard (**figure C**).

With a band saw and drum sander, he gives the fingerboard some final shaping to complete the design (**figure D**).

Figure C

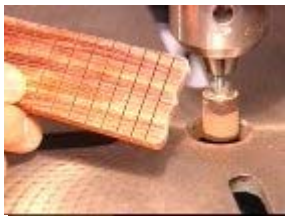
Figure D

Now, like the mandolin's body, the fingerboard receives some decorative binding around the outside edge (**figure E**). Just like the body, it also receives two layers with the tortoise shell binding on the outside.

Using a drill press, Lynn cuts shallow holes into the fingerboard where position markers will be inlayed. The location of each marker is determined by traditional placement, following the pattern of a vintage F5 (**figure F**).

Figure E

Figure F

Using some fine detail work, Lynn then actually sculpts thin chips of black Tahitian mother-of-pearl until they fit into a brass tube (**figure G**). Then he slices the tube to create a "button" outlined with brass.

With the buttons created, he inserts an inlay into each cavity using a filler of epoxy and cocobolo dust (**figure H**).

Figure G

Figure H

This part of the process gets fairly messy, but after a thorough sanding job with a radiused sanding block Lynn has completed a beautiful inlayed fingerboard (**figure I**).

Figure I

Creating the Peg-Head Veneer

The other preliminary detail Lynn fashions for the neck is a peg-head veneer. For that, he goes back to the CNC (computer numerical control) machine used in earlier segments and takes advantage of its precise cutting ability. Once again, with programming information supplied by Lynn, and based on classic patterns, the computerized router cuts a groove into the cocobolo (**figure J**). It also cuts the holes required for tuners in the exact locations Lynn has specified.

Figure J

The bit doesn't actually cut out the peg-head shape from the veneer. It leaves a channel that makes binding easier.

Lynn adds decorative binding to the veneer in two layers along the channel created by the CNC (**figure K**).

Using a chisel and furniture scraper, he then brings the profile of the binding down so that it's even with the surface of the cocobolo (**figure L**).

Figure K

Figure L

At that point he just pops out the peg head veneer from the blank (**figure M**). With the fingerboard and peg-head veneer completed, he has essentially finished the first preliminary steps of building a neck for the mandolin.

In the segment that follows, the neck itself is carved from a solid block of mahogany and made ready to be joined to the mandolin body.

Figure M

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HANDMADE MANDOLIN, PART 8: COMPLETING THE NECK

From "Handmade Music"
episode DHMM-106 -- More Projects »

In this final segment in this episode of DIY's *Handmade Music*, the neck of the mandolin is carved from a solid block of curly-maple and made ready to be joined to the mandolin body.

A newly made Dudenbostel mandolin gets the seal of approval from Kentucky Thunder's mandolin virtuoso Andy Leftwich.

Materials:

Curly-maple stock
Band saw
Table saw
Instrument-maker's glue
Hand-tools (chisels, scrapers, finger-planes, etc.)
Straight-edge
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Completing the Neck and Dovetail Joinery

- The stock used to create the neck itself is a 4x4 piece of curly maple (**figure A**).
- With the help of a template, Lynn cuts a rough shape for a mandolin neck using the band saw. Then, again using the band saw, he slices the neck in half lengthwise (**figure B**). He does this so he can sandwich a decorative slice of cocobolo wood down the middle of the instrument's neck.

Figure A

Figure B

- Using epoxy, Lynn combines the three pieces of wood (**figure C**).
- After a couple of hours of dry time, he cuts away the extra cocobolo leaving the basic form for the neck (**figure D**).

Figure C

Figure D

- Later, he'll install a **truss rod** in the mandolin's neck. A truss rod (**figure E**) is a metal rod that's installed in the neck that allows for adjustment to counter the force of wood movement caused by the tension of the strings. For new, Lynn cuts the channel for the truss rod at the table saw, and checks to make sure the channel is the proper depth (**figure F**) He'll install the rod later.

Figure E

Figure F

As seen in comparison template for the peg head, the stock for the neck isn't as wide as the finished peg-head (**figure G**).

To provide the extra width necessary, Lynn glues two lateral wooden extensions -- or "ears" -- at the location of the peg-head (**figure H**). "To make a mandolin neck out of a single piece of wood," says Dudenbostel, "would be very wasteful because you need such a wide piece of wood for the peg-head of the instrument. So, traditionally what's been done is to select a piece of wood about 1-3/4" wide for the main part of the neck. Small ears glued onto the side of the peg-head give you that width that you need."

Figure G

Figure H

Earlier Lynn cut the dovetail into the body of the mandolin (**figure I**). Now he has to prepare the matching cut for the neck.

Lynn does that by placing a strip of paper under the instrument and tracing the dovetail's shape. He then cuts out the dovetail on paper to create a template. He double-checks his template by on the instrument. If this paper template fits precisely in the dovetail (**figure J**), he's assured of getting a good bond between the neck and body.

Figure I

Figure J

With the template lying on the neck, Lynn transfers the dovetail pattern from paper to the neck's base. He then cuts the matching piece of the dovetail at the band saw (**figure K**).

After a rough-cut on the band saw, he does the final fitting of the dovetail by hand. He uses hand tools such as chisels and sanding sticks to finalize the dovetail (**figure L**) for a perfect fit.

Figure K

Figure L

- He then checks the fit of the dovetail by dry-fitting the head and neck-piece together (**figure M**). "Power tools only get you so far," Lynn says of this part of the process, "Then you have to finish up by hand. I'll take chisels and small sanding sticks and go in and finish the dovetail to get a very tight fit when the neck slides into the body. You want a very strong joint right there.
- Next Lynn can glue the peg head veneer to the ears he prepared earlier. First he attaches the front veneer using instrument maker's glue (**figure N**). After drying he uses instrument makers glue again to bond the back veneer which is also cut from cocobolo.

- With a band saw, Lynn then cuts away the extra maple from the peg-head (**figure O**).

Figure N

Figure O

- Lynn can now begin the process of carving the neck to its final shape. First, with the fingerboard in place, he marks the precise location of the fingerboard onto the neck piece (**figure P**). Based on these markings, he rough-cuts the neck's shape at the band saw.

- Then he begins using a variety of carving tools to begin shaping the neck (**figure Q**). This part of the process takes several hours, divided about evenly between carving out the basic shape and fine-tuning the exact shape and surface of the neck.

Figure P

Figure Q

In the episode that follows, Lynn finalizes the shaping of the neck then assembles and finishes the mandolin.

Luthier Lynn Dudenbostel gets satisfaction by hearing his instruments played in the hands of skilled musicians. According to Lynn, he can pick out the sound of an instrument he's made -- much like a parent can pick out the voice of his or her child among a group of children.

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Kentucky Thunder
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GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

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- Handmade Mandolin, Part 8: Completing the Neck

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HANDMADE MANDOLIN, PART 9: NECK, FINGERBOARD AND PEG-HEAD

From "Handmade Music"
episode DHMM-107 -- More Projects »

Thus far in the mandolin series, we've learned a great deal from one of America's great luthiers, Lynn Dudenbostel. With his guidance, we've learned the finer points of hand-building mandolins following in the traditions of the master instrument-builders of the 1920s and '30s.

In this episode of DIY's *Handmade Music*, viewers see the final steps in the building of a mandolin as Lynn joins the neck with the body of the instrument and adds the incredibly intricate inlay to the peg head. In the concluding segments, Lynn reveals "trade secrets" that are keys to creating a perfect finish. Finally, we find out just how good the mandolin sounds when the bluegrass band Kentucky Thunder christens the instrument in a Nashville recording studio.

In this first segment, Lynn completes the carving of the neck, adds the fingerboard and creates the inlay for the peg-head.

Luthier Lynn Dudenbostel enters the final stages in the building of a handmade mandolin based on the classic Gibson F5 model. In this segment, the final details are added to the neck, fingerboard and peg-head. The neck is then joined to the instrument's body using an precision-carved dovetail joint.

Andy Leftwich, mandolin virtuoso with the band Kentucky Thunder, samples the tone of a newly made Dudenbostel mandolin.

Materials:

Band saw
Drill press with sanding attachment
Woodworker's hand-tools -- files, rasps, hand scrapers, etc.
Sandpaper
Sanding block
Mother of pearl
Jeweler's saw
Forceps
Clamps
Instrument-maker's hide glue
Epoxy glue
Cocobolo-wood dust
Carpenter's pencil
Safety glasses or goggles

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Neck, Fingerboard and Peg-Head

So far, Lynn has built the mandolin's body and partially completed neck. In the previous segment, Lynn explained that ninety percent of the neck-carving can be done in about three hours, but that last ten percent, with it's fine detail work, could take quite a bit more time. That's the stage the Lynn now undertakes as the neck begins to take on its final shape (**figures A and B**).

When building an instrument, the musician is always on Lynn's mind so, while calipers and measurements can be used, nothing lets him know he's finished more than his own hand grasping the neck. When it feels right and feels like it's ready to be played, Lynn is finished carving.

Figure A

Figure B

With the carving of the neck complete, Lynn takes care of the finer details. Sanding will brings the neck closer to perfection. He does the major of the sanding on the drill press (**figure C**), a process that takes Lynn only about 15 minutes.

He follows this up with detailed hand-sanding (**figure D**), which may take a couple of hours.

Figure C

Figure D

Lynn's attention now turns to the **peg head**. To accept the mandolin's tuners that will be installed later, holes were pre-drilled earlier in the veneer on the face of the peg head. Now, using the drill press, Lynn drills the holes the rest of the way through the peg head (**figure E**) -- i.e., through the maple stock and the back veneer.

Figure E

At this point, Lynn goes back to the mandolin's body and dry-fits the neck to the body. If necessary, he can make minor adjustments to the dovetail, or add tiny wood shims, to ensure a tight, seamless fit. Once he's determined that the fit is good, he applies hot hide-glue to the faces of the dovetail on the body section (**figure F**) and to the heel of the neck piece. (Hide-glue, which is made from ground animal hides, is ideal for soft-wood applications.)

Figure F

As soon as the glue is applied, Lynn joins the neck the body (**figures G and H**). The pieces are clamped carefully and left to dry overnight.

Figure G

Figure H

With the neck and the body joined there's still one critical piece left to make before attaching the fingerboard to the neck -- the **fingerboard extension** which is characteristic of the F5-style mandolin. This raised piece is placed on the mandolin top, directly onto the spruce (**figure I**), where it butts against the 15th-fret cross-piece and supports the fingerboard. Lynn makes this piece from a piece of maple and glues it to the end of the fingerboard. "This was one of the big developments of the F5 in 1922," says Lynn, "to have a fingerboard extension that wasn't glued directly to the top of the instrument. That gives more surface area, that's free to vibrate, to the top of the instrument, and gives a different sound."

Figure I

Once the extension has been added, and the glue has dried, Lynn uses a sanding block to level the top surface of the extension so that its perfectly flush with the support for the fingerboard (**figure J**). "I try to leave just a few thousandths of an inch high," says Lynn, "so that I can sand it down to the plane of the fingerboard."

- With the extension glued in place and sanded flush, the next step is to install the fingerboard. Lynn lines up the fingerboard in dry-fit (**figure K**), and checks the alignment very carefully (**figure L**). Lynn is emphatic about conducting meticulous dry-fits before gluing. "Always fit everything before you glue it," he says. "Once you put glue on and assemble some of these joints, such as a dovetail, even just a few seconds after you slide it together, it's near impossible to take it apart."

- Once the fingerboard is glued in place, it's left to dry overnight.

Figure J

Figure K

Figure L

Inlay for the Peg-Head

Next, Lynn begins work on the ornate inlay for the mandolin's peg head.

- In some cases, the mandolin's inlay helps with identification. In the case of a Dudenbostel mandolin, Lynn inserts mother-of-pearl cutouts of his name into the instrument's peg head (**figure M**). He first cuts the channel using a rotary tool, then places the inlay in the channel using a mix of epoxy and cocobolo dust.

- Once glue has dried, Lynn sands the surface using 150-grit sandpaper. Once sanded flush with the wood, the inlaid letters really stand out. This effect is appears especially dramatic when the wood surface gets darkened with a damp rag that's used to remove the excess dust (**figure N**).

Figure M

Figure N

- Next Lynn focuses on the decorative figure portions of the inlay. He copies a template pattern (**figure O**) onto various decorative shells, and cuts out the shapes using a jeweler's saw. He also cuts the rest of the pattern's material and traces the entire shape onto the peg head.

- Using a rotary tool, he cuts away a channel to embed the pattern. He then places the cut inlays into the channels (**figure P**). His experience with inlay usually guarantees a perfect fit for the design but his epoxy mixture fills any gaps.

- After sanding away the dry mix Lynn is left with a peg head featuring a beautiful inlay pattern (**figure Q**). For those who admire quality instruments, this is a striking example of fine workmanship.

Figure O

Figure P

With the neck completed and joined to the body, the instrument is termed "in the white" -- which is a luthier's way of saying everything is complete except for the finish. In the segment that follows, the mandolin's finishing process gets underway with applications of stain and shellac.

RESOURCES:

Kentucky Thunder
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Figure Q

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

Mandolins are certainly judged by their sound, but their look can indicate quality too. The decorative inlay on an instrument can be important to a luthier's reputation.

- **ALSO IN THIS EPISODE:**
- Handmade Mandolin, Part 9: Neck, Fingerboard and Peg-Head
- Handmade Mandolin, Part 10: Wood Stain and Shellac
- Handmade Mandolin, Part 11: Final Wood Finish
- Handmade Mandolin, Part 12: Setup and Finishing Touches
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HANDMADE MANDOLIN, PART 10: WOOD STAIN AND SHELLAC

From "Handmade Music"
episode DHMM-107 -- More Projects »

With the neck completed and joined to the body, the instrument is termed "in the white" -- which is a luthier's way of saying everything is complete except for the finish. Alhough Lynn Dudenbostel has completed the construction on the handmade mandolin, it is not ready to be heard. The wood finish does more than add to the look. It's also an important aspect of an instrument's sound.

In this segment, the mandolin's finishing process gets underway with applications of stain and shellac.

Lynn chose a honey-colored stain for this mandolin. Staining is only the first step in finishing process. Staining is followed by the application of shellac and varnish. The final step is a technique known as a French polish.

Materials:

Spray booth and paint sprayer
Stain
Shellac
Varnish
Paintbrushes
Rags
Sandpaper
Masking tape
Protective gloves

Safety Alert: When working with stains, varnishes, solvents and other toxic chemicals, work in a well-ventilated area. For some procedures, a ventilator mask is recommended.

Kentucky Thunder mandolin-player Andy Leftwich plays one of Lynn Dudenbostel's custom handmade mandolins.

Wood Finish, Part 1: Wood Stain and Shellac

- Lynn preps the mandolin for the finish by raising the grain of the wood. He applies a small amount of water to the wood surface and carefully sands using 150-grit sandpaper (**figure A**). He repeats the process several times, using progressively finer grit sandpaper. For the final sanding he uses 220-grit sandpaper.

- He then masks off the cocobolo-wood fingerboard (**figure B**) and peg head with masking tape to protect the detail work during the staining process.

Figure A

Figure B

- Rather than staining this mandolin in the traditional sunburst style of the original F5 mandolins of the 1920s, Lynn opts for a distinctive honey-colored finish for this mandolin. He applies a light, amber stain to the surface of the mandolin using a rag (**figure C**). A small amount of potassium dichromate

is also used to give the instrument a vintage or aged look. Lynn applies a even coat of the stain and allows it to dry overnight.

Lynn assesses the look of the wood after each coat and decides whether to add more. In this case a second coat of stain is sprayed on (**figure D**).

Figure C

Figure D

Staining is only the first step in the finishing process. After the stain has been applied, and has dried fully, the wood is sealed with shellac. Lynn applies the shellac, mixed with a small amount of the stain, using a sprayer (**figures E and F**). "The shellac acts as a binder between the wood and the varnish," he says. "That's kind of important to have that transition there. The varnish doesn't stick as well to bare wood as it does to shellac."

Figure E

Figure F

A couple of coats of coats of shellac are applied and allowed to dry, then the mandolin will be ready for the next step in the finishing process.

In the segment that follows varnish is applied, and the finish is finalized using an old-school technique known as French polishing.

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HANDMADE MANDOLIN, PART 11: FINAL WOOD FINISH

From "Handmade Music"
episode DHMM-107 -- More Projects »

Building a mandolin is a major undertaking but, at this point when the varnish is applied, the payoff becomes apparent. With a number of coats of stain and shellac applied and fully dry, the mandolin is now ready for the next step in the finishing process. In this segment, the varnish is applied and the finish is finalized using an old-school technique known as French polishing.

Lynn Dudenbostel applies the varnish by hand, using a brush rather than a spray booth.

Materials:

Varnish
Paintbrushes
Rags
Sandpaper
Masking tape
Protective gloves

Safety Alert: When working with stains, varnishes, solvents and other toxic chemicals, work in a well-ventilated area. For some procedures, a ventilator mask is recommended.

Lynn may apply as many as eight coats of varnish, but sands away much of that between coats.

Wood Finish, Part 2: Varnish and French Polish

Lynn applies the varnish using a brush (**figures A and B**) rather than a spray booth. It's still the best method of application. "In brushing the varnish," says Lynn, "I find that it takes anywhere from six to eight coats to accomplish a proper finish." But as Lynn is fond of saying, it's not how many coats are *added* to the instrument, it's how many are *left*.

Figure A

Figure B

Although he might add as many as eight coats of varnish, it's important to note that he also sands away much of that between applications. "I'll let it set for 24 hours," he says, "then come back with 800-grit sandpaper and very lightly scuff that coat of varnish that I've put on." Following this technique he can achieve a remarkably smooth and attractive finish that's thin enough not to hamper the sound of the instrument.

Lynn begins slowly, adding coats only as necessary. Each coat applied is followed by sanding. The initial between-coat sandings are dry, but for the later coats Lynn does wet-sanding using ultra-fine 1500-grit sandpaper (**figure C and D**).

Figure C

Figure D

After wet sanding, depending on what the finish "tells" Lynn, there could be more coats of varnish applied (**figure E**), followed by more wet sanding. Ultimately he finishes with 1500-grit paper. A light touch is required for sanding since he never wants to sand through to bare wood. It's a balancing act, getting enough on to achieve a smooth, perfect surface, but not leavening any more on than is necessary so the acoustics aren't compromised. By the end of this ritual, the wood has taken on a beautiful luster.

Figure E

And after one more light-sanding Lynn's ready for the final finishing step -- **French polishing**. "To get a nice gloss back on the finish," says Lynn, "you French polish it [which is] little more than shellac, with a little bit of walnut oil as a lubricant, that's applied with a pad. This is the way the instrument in the '20s were done -- with an oil-based varnish with a spirit top-coat. It adds just a little bit of shellac to the top layer of varnish, bringing it to a nice shine." (**figure F**).

Figure F

With the completion of the French polish, the mandolin is complete with the exception of set-up and stringing. Those steps are covered in the next, final segment in the mandolin series.

RESOURCES:

Kentucky Thunder
For more information on the band that appeared in DIY's *Handmade Music*, Kentucky Thunder -- and to hear streaming audio and download samples of their music -- visit the website for Skaggs Family Records, www.skaggsfamilyrecords.com.

Stewart MacDonald's Luthier Supply Shop
Website: www.stewmac.com

Luthier's Mercantile International, Inc.
Website: www.lmii.com

Pioneer Valley Luthier Supply Company
Website: www.pioneervalleyluthier.com

GUESTS:

Lynn Dudenbostel, Luthier
Dudenbostel Stringed Instruments
Knoxville, TN

- **ALSO IN THIS EPISODE:**
- Handmade Mandolin, Part 9: Neck, Fingerboard and Peg-Head
- Handmade Mandolin, Part 10: Wood Stain and Shellac
- Handmade Mandolin, Part 11: Final Wood Finish
- Handmade Mandolin, Part 12: Setup and Finishing Touches
- WEB EXTRA: *Handmade Music* Song Downloads

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HANDMADE MANDOLIN, PART 12: SETUP AND FINISHING TOUCHES

From "Handmade Music"
episode DHMM-107 -- More Projects »

With the completion of the French polish, the mandolin is complete with the exception of set-up and stringing. Those steps are covered in this final segment in the handmade mandolin series.

The maker's mark. Lynn Dudenbostel signs his name to the plate that's mounted inside the instrument indicating who made the instrument and the date of completion.

Materials:

Mandolin tuners
Mandolin strings
Hand saw
Tap-hammer

Safety Alert: *Always* wear safety goggles or safety glasses when working with wood, power-tools, saws, drills, routers, etc.

Set-Up and Finishing Touches

- Lynn's first step in stringing -- or setting up -- the mandolin is to clean the fret slots and add bushings to the tuner holes. Fret wire is cut and shaped to fit each individual fret-slot on the fingerboard (**figures A and B**).

Figure A

Figure B

- They are tap-hammered into place and the fit is secured without glue (**figure C**).
- Each fret is trimmed along the fingerboard so a player's hand can glide along the edge without snagging on protruding metal (**figure C**). With 22 frets installed, trimmed and filed, the fingerboard is complete.

Figure C

Figure D

- Before the mandolin can be strung up the tuners have to be added. The tuners are inistalled in the pre-drilled holes in the peg head (**figure E**) and are fastened with matching brass screws (**figure F**). Cocobolo buttons on the tuners were added to match the rest of the mandolin.
- The inlayed cocobolo tailpiece adds a decorative touch but it's also vital because the strings are hooked inside this base. And then they'll cross over the bridge.

Figure E

Figure F

The cocobolo bridge is positioned precisely (**figure G**) but isn't held in place with glue. Instead it essentially "rides" on the top under the pressure of the strings.

Lynn notches gaps into the edge of the bridge to allow the strings a smooth path on the way to the tuners.

Figure G

Likewise the nut, made of fossilized ivory, must be notched (**figures H and I**) because it too guides the strings and keeps them in their proper position. The position is critical, but another consideration for both the bridge and the nut is the depth of the gaps.

Figure H

Figure I

The adjustment on the depth of the string channels (**figures J and K**) will affect how high the strings rise above the fingerboard. Put another way, the depth effects the **action** of the strings. Action is purely a personal preference so Lynn has to communicate with his customers regarding what they want from their instrument. Delivering the right action can take time.

Figure J

Figure K

With the strings properly located, Lynn is on the home stretch. He makes final fine-adjustments to the strings (**figure L**), bridge and nut.

The mandolin's tuners.

Figure L

One of the last steps is the installation of the pick guard (**figure M**). With those final steps, a new mandolin has been born (**figure N**). In putting these finishing touches on this instrument, Lynn completes his thirtieth handmade mandolin. But that kind of output has by no means bred apathy. "Building instruments is just a great joy," Lynn says. "I've often said if I never sold another one, I guess I'd have a house full of them. I would just continue to build. It's a great sense of satisfaction building an instrument all the way through to completion. I've built about 75 instruments at this point, and still, to this day I have that same feeling of excitement when I string up an instrument for the first time. It's just like that very first one I built. To be able to take a pile of raw materials that look like not much more than firewood, and transform them into something that not only looks beautiful, but sounds wonderful also, is just one of the greatest experiences you can have."

Figure M

Figure N

Recap: Dudenbostel Handmade Mandolin

- As Lynn said, we started with little more than scraps of firewood. More specifically, we started with two prime pieces of West Virginia Adirondack red-spruce and joined them to form a top.
- That top and a solid piece of curly maple were carved into the shape of a mandolin with the aid of a computerized router.
- High tech yielded to low tech with the use of finger planes making the final graduations to give the top and back perfect tone.
- The sound was accentuated with perfectly carved tone bars. The purpose of the tone bars is to create an environment inside the instrument where air can create vibrations or sound waves that emit perfect sound.
- The top and back needed to meet with sides and we created a rim of maple by bending strips around a heated metal pipe.
- With all those pieces joined and decorative binding added Lynn completed the construction of the mandolin body.
- Neck construction was a challenge but Lynn was up to the task. Cut from four by four stock and carved over several hours with a variety of tools Lynn added a cocobolo fingerboard and attached it all to the body with a dovetail joint and hide glue.
- The finish was achieved with a mix of stain, shellac and varnish and created a non-traditional take on a vintage style instrument that expresses the creativity of its maker.

Lynn Dudenbostel says that, after having built more than 75 stringed instruments, he still gets a sense of satisfaction and excitement each time he strings up a new one and hears it play its first notes.

Mandolin virtuoso Andy Leftwich gives enthusiastic approval to the sound of the newly made Dudenbosel mandolin.

Andy prefers a mandolin with a lower action (i.e., string height), making it easier to play faster, although there is some trade-off with the volume and projection of the sound delivered by the instrument.

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